MILITARY SPECIFICATION SHEET

RELAYS, HYBRID, TIME DELAY (ON OPERATE), CLASS B, TYPE I, HERMETICALLY SEALED, 4PDT, 10 AMPERES, ADJUSTABLE TIME DELAY (EXTERNAL RESISTOR), 0.1 TO 600 SECONDS

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83726.

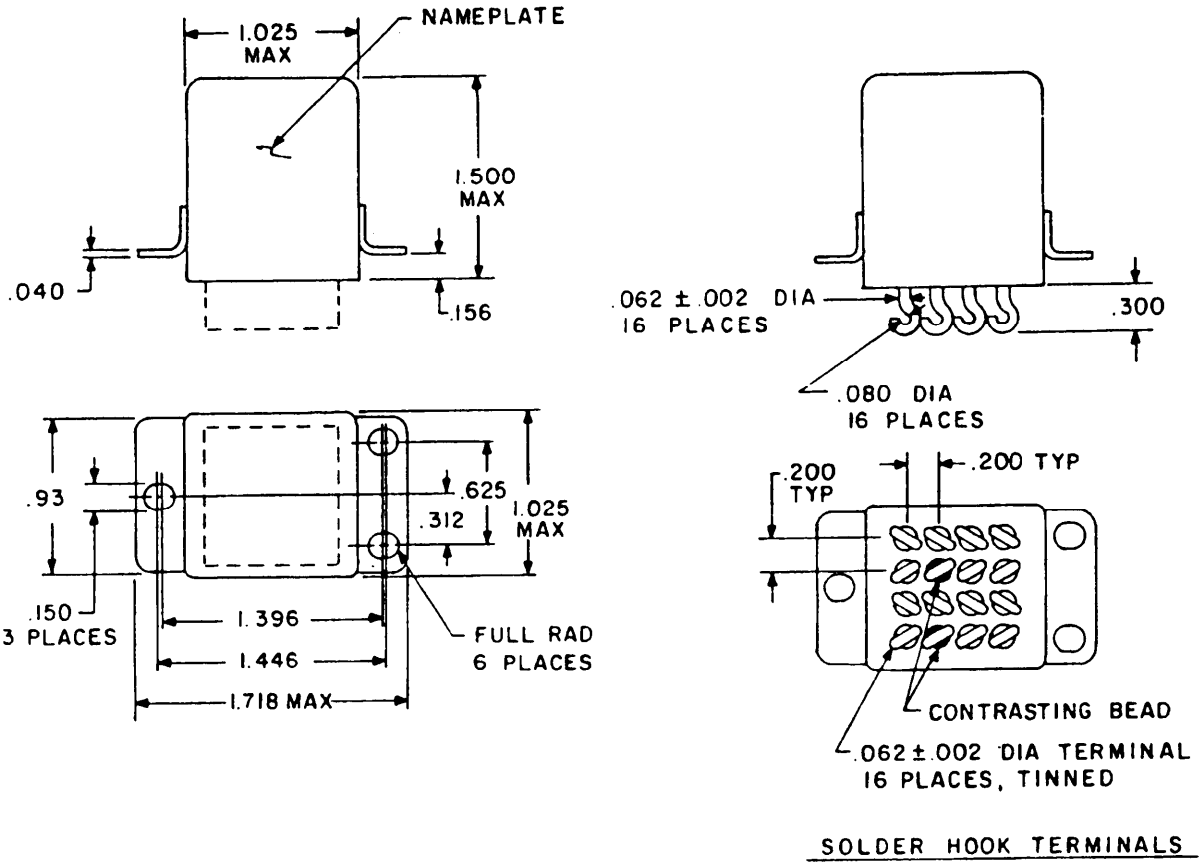
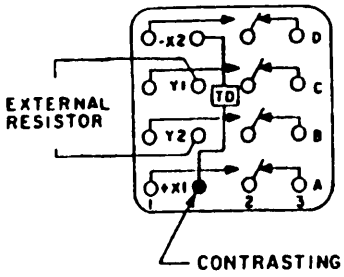


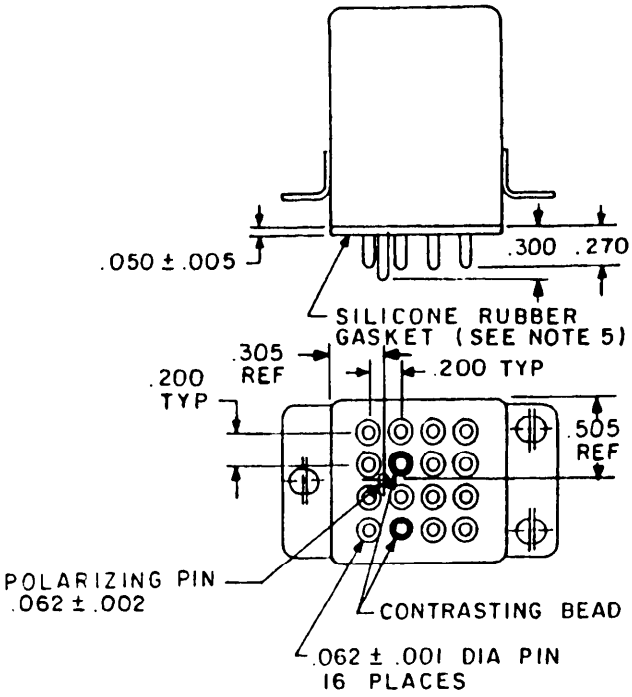
FIGURE 1. Outline dimensions and configuration of relay.

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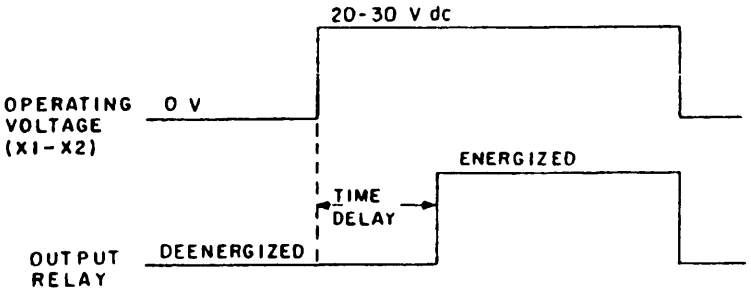
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CIRCUIT DIAGRAM (SEE NOTES 4 AND 7)
TERMINAL VIEW



PLUG-IN TERMINALS
(SEE NOTE 6)



TIMING DIAGRAM

Inches	mm
.040	1.02
.050	1.27
.062	1.57
.080	2.03
.150	3.81
.156	3.96
.200	5.08
.270	6.86
.300	7.62
.312	7.92
.625	15.78
.937	23.80
1.025	26.04
1.396	36.73
1.446	36.73
1.500	37.10
1.718	43.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
4. Terminal numbers shall not appear on the relay header and there shall be a legible circuit diagram on relay which identifies each terminal location specified.
5. Gasket material: Silicone rubber gasket AMS3332. Shore hardness 20 ±5.
6. Terminal plating: Gold in accordance with MIL-G-45204 type II, class 1. Underplating, nickel 50 to 150 microinches thick.
7. External resistor, in accordance with MIL-R-55182 (RNC60HXXXXFS), is selected as follows:

$$R_{EXT} = \left(\frac{T_1}{T_0} - 1 \right) 100\ k$$

T_0 = Minimum time: lower limit of decade range (see part number requirement).
T_1 = Required time
$T_1 \leq 10\ T_0$

FIGURE 1. Outline dimensions and configuration of relay - Continued.

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REQUIREMENTS:

OPERATING DATA:

Timing action: Delay-on operate.

Time delay: Adjustable; select decade range within 0.1 to 600 seconds.

Timing accuracy 1/: ±10 percent of nominal value.

Recycle time 2/: 50 milliseconds maximum.

Power interrupt 3/: 500 microseconds.

Operate current: 150 milliamperes maximum at 25°C.

INPUT DATA:

Input voltage 4/: 20 to 30 V dc.

Duty rating: Continuous.

Polarity protection: The timer shall be inoperative during, and undamaged by, reversal of the polarity of the operating voltage.

OUTPUT DATA:

Configuration: 4PDT.

Life: See table I.

TABLE I. Life load ratings (relay case grounded).

Type of load	Life (cycles)	Amperes 28 V dc	Amperes 115/200 V ac 400 Hz
Resistive	100,000	10	10
Inductive	20,000	8	8
Motor	100,000	4	4
Lamp	100,000	2	2
Low level	100,000	Applicable in accordance with MIL-R-6106	---

1/ The accuracy requirement applies for any combination of operating temperature and voltage. External resistors shall be in accordance with MIL-R-55182, characteristic C or equivalent. Add ±10 milliseconds for timing less than 1 second.

2/ Recycle time is defined as the minimum time that power must be removed from the input terminals to assure that the next timing cycle will be completed within the specified timing tolerance. (Units can be recycled during timing or after time-out).

3/ Transient and power loss specifications are based on a maximum duty cycle of 1/50.

4/ Minimum high temperature test 21 V dc; minimum continuous current test 23.5 V dc.

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ELECTRICAL DATA:

Transients: +80 volts, in accordance with MIL-STD-704, figure 9, limit 1.

Spike:

Self-generated: ±50 volts maximum.

Spike transients: ±600 volts, 10 microseconds maximum.

Susceptibility: +80 volts maximum, -600 volts maximum.

Electromagnetic interference 5/: In accordance with MIL-STD-461, class 1D.

Insulation resistance 6/: 1,000 megohms at 500 V dc at sea level, and 100 V dc at 80,000 feet between each pin and case.

Dielectric withstanding voltage 6/: 1,000 V rms at 60 Hz at sea level, and 350 V rms at 80,000 feet between case and pins connected together.

ENVIRONMENTAL DATA:

Ambient temperature:

Operating: -55°C to +125°C.

Nonoperating: -65°C to +125°C.

Vibration:

Sinusoidal: 30 G, 10 to 3,000 Hz.

Random: 0.4 G^2/Hz power spectral density, 50 to 2,000 Hz in accordance with MIL-STD-810.

Shock (specified pulse): 100 G, 6 ±1 milliseconds, 1/2 sine, 3 axes.

Acceleration: 15 G in any axis.

Seal: Hermetic.

PHYSICAL DATA:

Dimensions and configuration: See figure 1.

Terminations: See figure 1.

Terminal strength: 3 pounds pull.

Weight: 3.0 ounces maximum.

Marking: See MIL-R-83726.

5/ EMI test limits will not be exceeded during the timing interval or when continuously energized under steady-state conditions in accordance with EMI test of MIL-R-83726.

6/ Terminals X1, X2, Y1, and Y2 must be connected together during this test. Dielectric withstanding voltage and insulation resistance are measured between all mutually insulated terminals and between all terminals and case.

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PART NUMBER 7/ 8/: Consists of the prefix, specification sheet number, a four-digit dash number (expressed in milliseconds), and a suffix letter from table II (indicating quality level) as follows:

M 83726/37- 1001 A

Prefix

Specification sheet number

The first three digits are significant. The fourth is the number of zeros to follow the first three digits (The time is expressed in milliseconds and converted to seconds.) See examples following table II.

Suffix letter from table II.

TABLE II. Suffix letters for quality levels.

Quality level indication (refer to MIL-R-83726)	P (plug-in)	S (solder-hook)
W	P	S
Y	A	B
X	C	D

Examples:

M83726/37-1001P: 0.1- to 1-second time delay, plug-in, W level.

M93726/37-9002B: 9- to 90-second time delay, solder hook, Y level.

M83726/37-6003C: 60- to 600-second time delay, plug-in, X level.

7/ Time delay relays within the 0.1- to 600-second delay range are available within a one-decade range. The part number represents the upper timing limit of each range.

8/ CAUTION: Relays supplied to this specification are susceptible to damage from electrostatic discharge under class I devices, as defined in DOD-STD-1686.

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
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Agent:
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(Project 5945-F637-6)

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